

AI Driver Drowsiness & Distraction Monitor

Project README / Setup and Usage Guide

This is a real-time computer vision system that keeps an eye on the driver and warns them when they start losing focus. It uses MediaPipe Face Mesh to track 468 3D facial landmarks, and from those it works out whether the driver is drowsy, yawning, or turning their head away from the road. Alerts escalate in three levels, and every session is logged so you can review it afterwards.

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Repository: github.com/murugavel03/AI-Driver-Drowsiness-Distraction-Monitor

Project Overview

Attribute	Details
Domain	Computer Vision / Real-Time Systems
Language	Python 3.8+
CV Library	MediaPipe (Google) + OpenCV 4.8
Detection	Eye Aspect Ratio, Mouth Aspect Ratio, Head Pose (PnP)
Alert System	Multi-level audio alerts via pygame
Logging	Per-frame CSV logging feeding post-session analytics
Training	None needed — uses pre-trained MediaPipe Face Mesh

What It Detects

Event	Method	Threshold
Eyes closing / drowsiness	Eye Aspect Ratio (EAR)	$EAR < 0.25$ for 0.5s
Microsleep	Consecutive low-EAR frames	longer than 2 seconds
Yawning	Mouth Aspect Ratio (MAR)	$MAR > 0.65$
Head nodding down	Head pitch angle	$> 20^\circ$
Looking away / distracted	Head yaw angle	$> 35^\circ$

Alert Levels

The alerts get louder and more insistent as the situation gets worse, rather than treating every event the same:

- Level 1 — CAUTION (yellow). The eyes are starting to close.
- Level 2 — WARNING (orange). Drowsiness is sustained, or a yawn was detected.
- Level 3 — DANGER (red). A microsleep. This is the emergency case.

Project Structure

```
driver-drowsiness-monitor/
├── README.md          # This file
├── requirements.txt    # Python dependencies
├── monitor.py         # Main real-time detection (run this)
├── analyze_session.py # Post-session analytics & charts
├── demo_simulate.py   # Demo without a webcam
├── utils/
│   ├── __init__.py
│   ├── face_metrics.py # EAR, MAR, head pose computations
│   ├── alert_system.py # Multi-level audio alert manager
│   ├── session_logger.py # CSV session logging
│   └── visualize_landmarks.py # MediaPipe landmark explorer
├── logs/              # Auto-created: session CSV logs
├── plots/             # Auto-created: analysis charts
└── screenshots/       # Auto-created: saved frames
```

Environment Setup

Before you start, make sure you have Python between 3.8 and 3.11, since MediaPipe will not install on anything newer. You will also need a webcam, or a video file if you want to test without one, and around 500 MB of free disk space.

Step 1 — Clone the repository

```
git clone https://github.com/murugavel03/AI-Driver-Drowsiness-Distracton-Monitor.git
cd AI-Driver-Drowsiness-Distracton-Monitor
```

Step 2 — Create a virtual environment

```
python -m venv venv

# Windows
venv\Scripts\activate

# macOS / Linux
source venv/bin/activate
```

Step 3 — Install the dependencies

```
pip install -r requirements.txt
```

If MediaPipe refuses to install, the version of Python is almost always the reason. Run `python --version` and check that it falls between 3.8 and 3.11.

Running the Project

There are a few ways to start it depending on what you have available:

```
# Option A: real-time webcam monitor
python monitor.py

# Option B: run against a video file instead
python monitor.py --source path/to/driving_video.mp4

# Option C: auto-calibrate the EAR threshold first (stare at the camera for 3s)
python monitor.py --calibrate

# Option D: set your own thresholds
python monitor.py --ear 0.22 --mar 0.70

# Option E: run without writing a session log
python monitor.py --no-log
```

All CLI arguments

Flag	Description
--source	Video source: 0 for webcam, 1 for a second camera, or a file path. Default is 0.
--ear	EAR drowsiness threshold. Default 0.25.
--mar	MAR yawn threshold. Default 0.65.
--calibrate	Auto-calibrate EAR from the live camera before starting.
--no-log	Turn off CSV session logging.

Keyboard shortcuts while running

Key	Action
q	Quit the monitor
s	Save a screenshot
r	Reset the session counters
c	Toggle the landmark mesh overlay

Post-Session Analysis

```
# Analyse the most recent session
python analyze_session.py

# Analyse one specific log
python analyze_session.py --log logs/session_20260912_103045.csv

# Analyse every session you have recorded
python analyze_session.py --all
```

This drops a report into plots/ covering EAR over time with blink events marked, MAR over time with yawn events, the three head-pose angles (pitch, yaw and roll), a pie chart of how the session was distributed across states, the alert-level timeline, and a block of summary statistics.

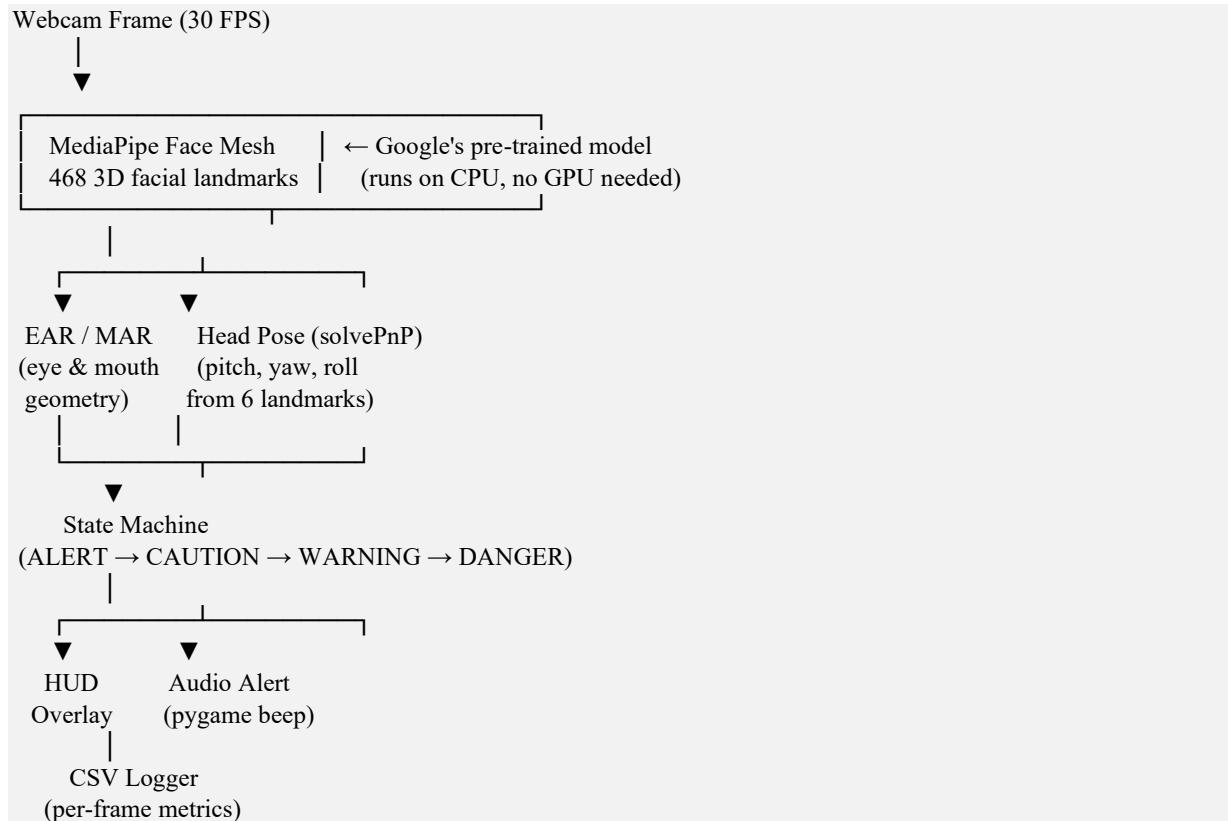
Demo Without a Webcam

```
python demo_simulate.py --duration 120 --plot
```

This generates a two-minute simulated session at logs/session_demo.csv and saves a chart into plots/. It exists mainly so the project can be graded on a machine that has no camera attached.

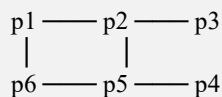
How It Works

The flow through one frame looks like this:



The Eye Aspect Ratio formula

$$EAR = (|p2-p6| + |p3-p5|) / (2 \times |p1-p4|)$$



Open eye → $EAR \approx 0.30$

Closed eye → $EAR \approx 0.05$

The gap between an open and a closed eye is wide enough that a single threshold works well in practice, which is what makes this approach worth using over something heavier.

Troubleshooting

Problem	Solution
ModuleNotFoundError: mediapipe	pip install mediapipe
No module named 'cv2'	pip install opencv-python
Webcam not found	Try --source 1 or --source 2
No audio alerts	pip install pygame
Low detection accuracy	Run with the --calibrate flag
MediaPipe install fails	Use Python 3.10: py -3.10 -m venv venv

References

1. Kartynnik, Y. et al. “Real-time Facial Surface Geometry from Monocular Video on Mobile GPUs.” CVPR Workshop, 2019. (MediaPipe Face Mesh)
2. Soukupová, T. and Čech, J. “Real-Time Eye Blink Detection using Facial Landmarks.” CVWW, 2016. (EAR for drowsiness)
3. OpenCV Documentation — solvePnP for head pose estimation.
4. Abtahi, S. et al. “YAWNNet: Detecting Yawning and Pose in the Wild.” 2014. (MAR for yawning)
5. National Highway Traffic Safety Administration (NHTSA). Drowsy Driving Statistics, 2017 Data.

License

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Developed as a Computer Vision course project for the VITyarthi Flipped Course Evaluation: an AI driver safety monitor built with MediaPipe and OpenCV.